Disorders of Perception

- https://upload.wikimedia.org/wikipedia/commons/1/1a/Change_blindness.gif
- https://www.cse.iitk.ac.in/users/se367/10/presentation_local/Change%20Blindness_files/plane.gif
- https://sites.socsci.uci.edu/~ddhoff/venice.gif
- https://upload.wikimedia.org/wikipedia/commons/7/7f/Change_blindness_illusion.gif

Visual Object Agnosia

- The impairment of object recognition in the presence of relatively intact elementary visual perception, memory, and general intellectual function.

Agnosia

- **Apperceptive**: lack of recognition because of impaired (higher order) visual perception.
- **Associative**: lack of recognition because of lack of association of percept with its meaning.

Apperceptive Agnosia

- Basic acuity and visual fields preserved:
  - Good acuity,
  - Brightness discrimination,
  - Color vision, and
  - Other elementary visual capabilities
- However, cannot distinguish between or recognize different visual shapes
- Cannot copy (draw) shapes

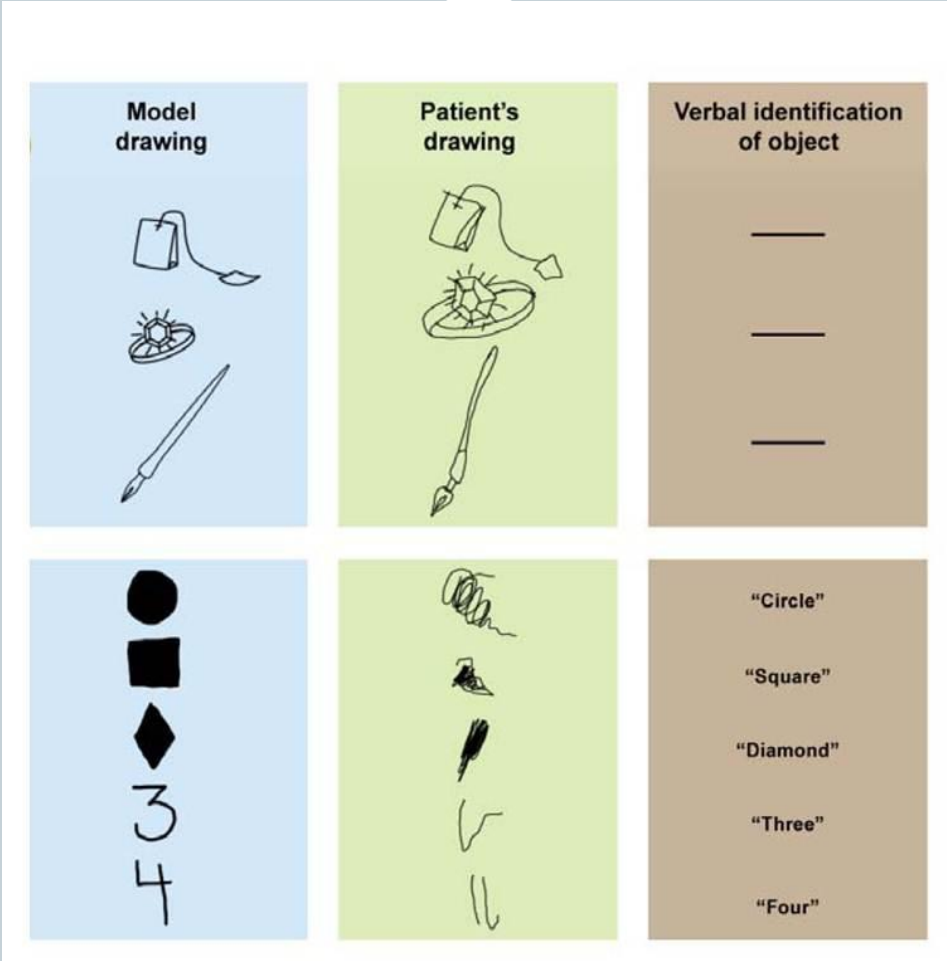
Apperceptive Agnosia

- Apperceptive agnosias often occur after carbon monoxide poisoning or other diffuse toxic effects.
- Subtypes include: dorsal and ventral simultanagnosia, and visual hemi-neglect.

Associative Agnosia

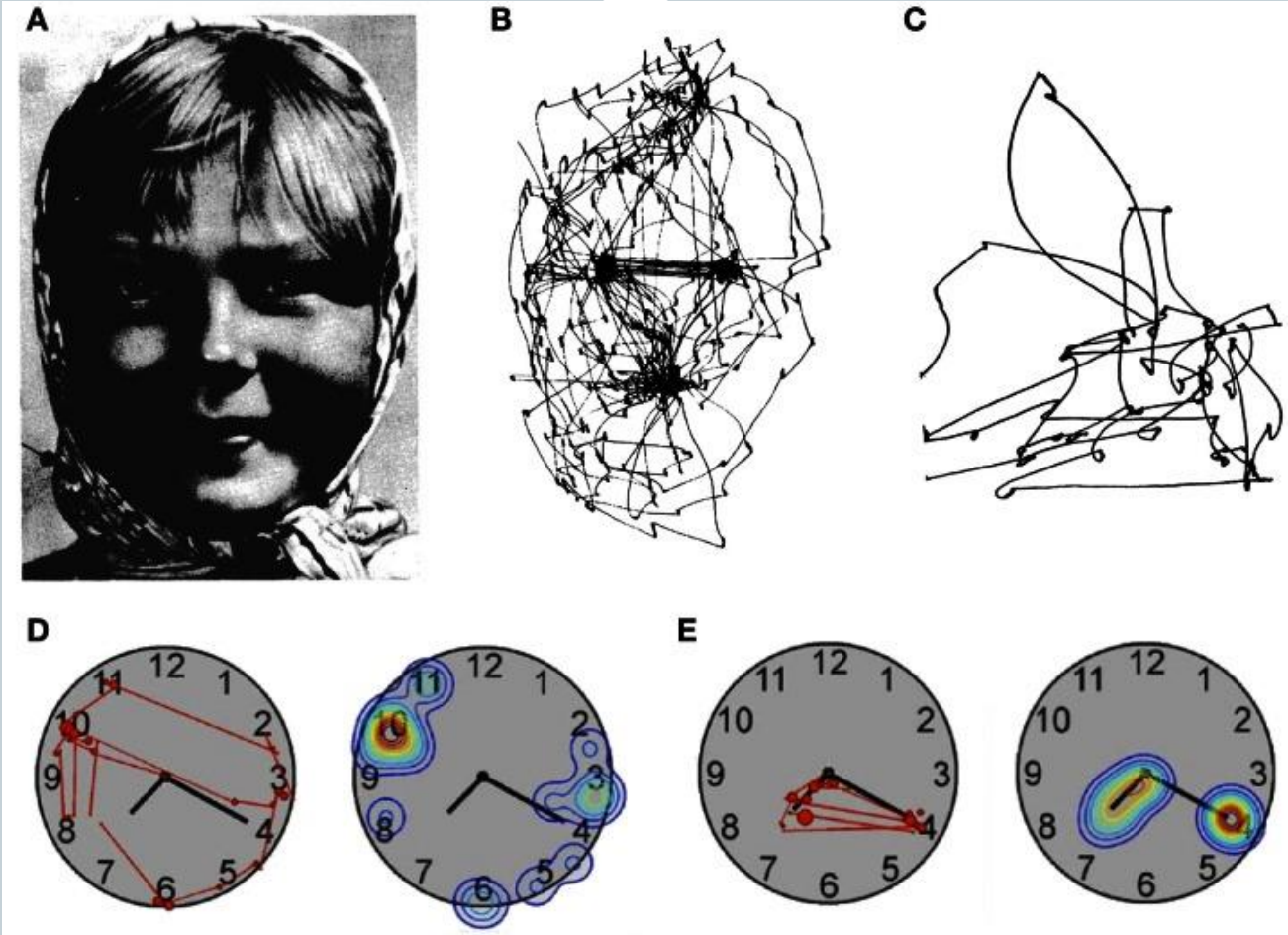
- Visual perception relatively preserved but not visual object recognition. “Recognition without meaning”
  - Can copy shapes (drawing)
  - Can recognize from verbal description or non-visual sense
  - Can recognize by feel or verbal description
  - Cannot indicate visual recognition by non-verbal means (Not the same as *Optic Aphasia,* which is the ability to name from description and palpation but a specific inability to name visually-presented objects.)
- Can be category specific (e.g. face, object, printed word)
- Generally occipitotemporal cortex + WM, sometimes L hemisphere or bilateral lesions.

top = associative agnosia
bottom = apperceptive agnosia



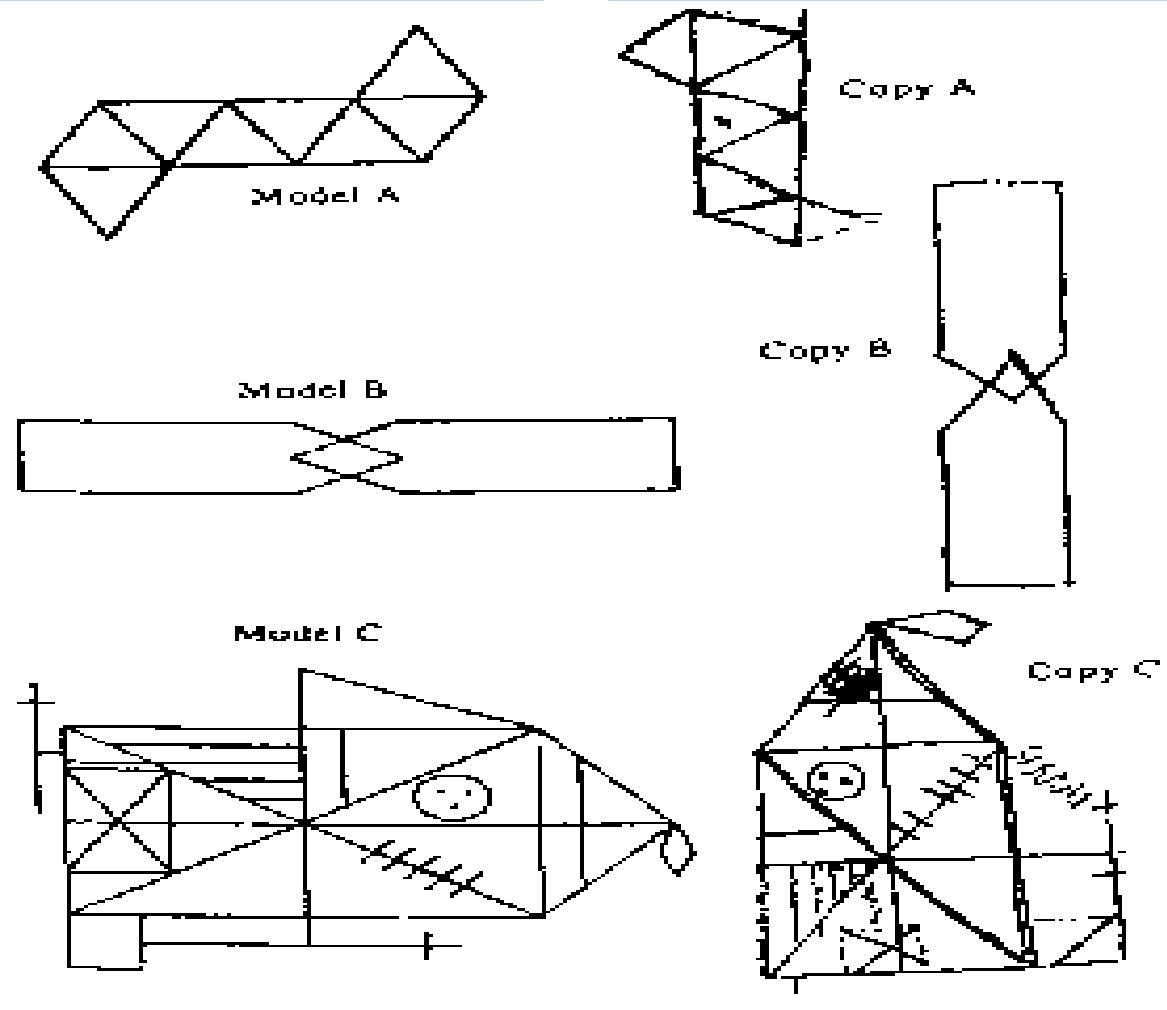
Simultanagnosia

- Impairment in perception of multielement or multipart visual displays. (e.g. piecemeal description of a complex picture. May fail to interpret overall scene.)
  - Dorsal: Cannot attend to ≥1 part of object. May act blind as they bump into objects that are close to each other. Have trouble counting.
    - Bilateral Posterior Parietal lesions.
  - Ventral: Recognition of simple shapes not obviously impaired. (letter by letter reading). Don't bump into objects i.e. don't act blind. Can count, so can actually 'see' more than one object. But don't see the whole picture in a scene, and can't identify more than one object at a time.
    - L inferior temporo-occipital cortex


A
B
C
D
E

Perceptual Categorization Deficit

- (Warrington and Taylor, 1973)
- Subjects have trouble matching 3D objects across differences in viewing conditions (changes in lighting, perspective).
- But no obvious problem in normal life: require changes in lighting etc. that even normals find challenging (Farah 1999). More a deficit in ‘visual problem solving’?
  - Poorly localizing, usually. R inferior parietal lobe.

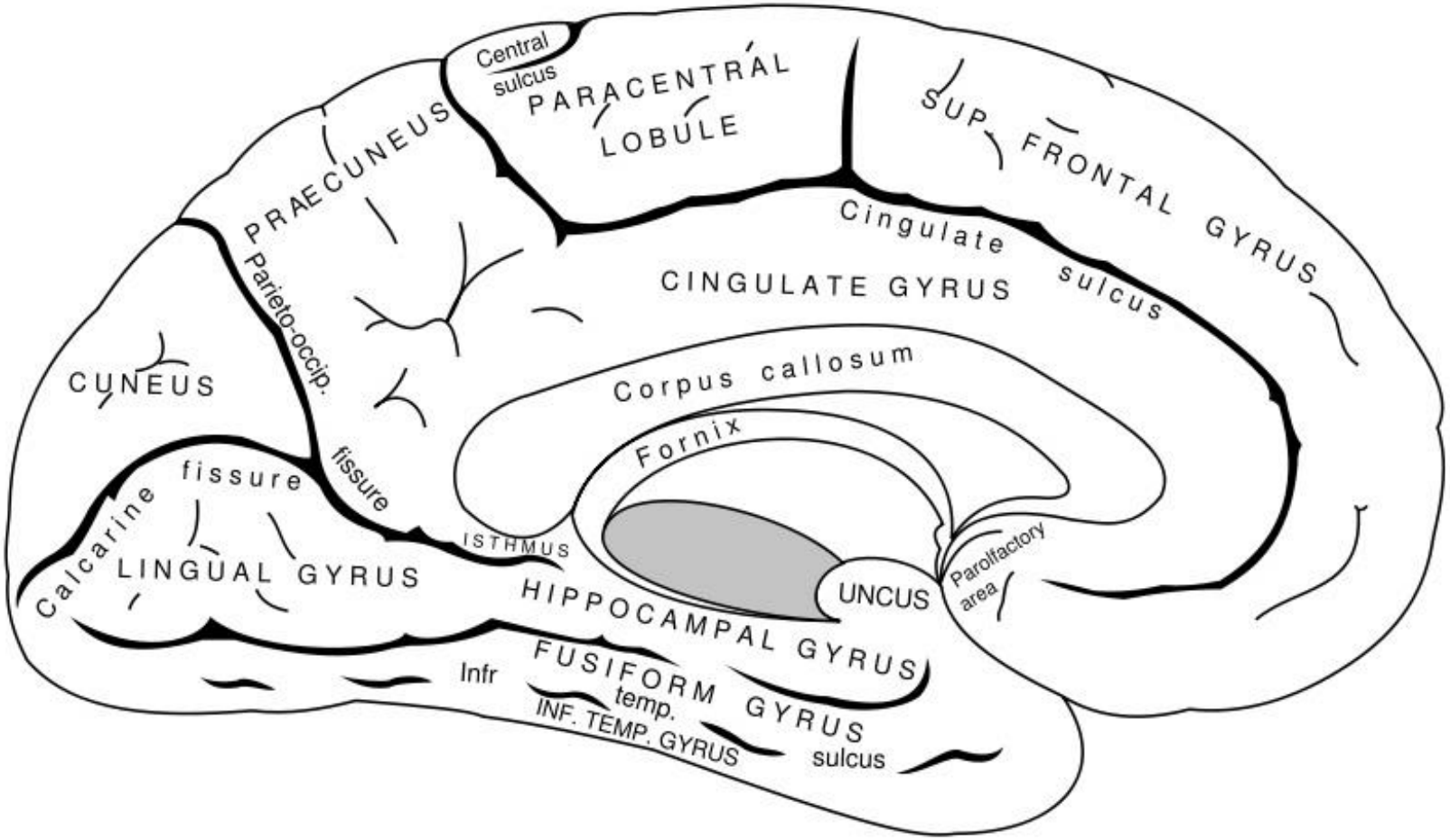

Model A
Copy A
Model B
Copy B
Model C
Copy C

Prosopagnosia

- “The man who mistook his wife for a hat”
- Προςοπόν (prosopon) = face, agnosia = non-knowledge.
- Usually aware of deficit.
- Sometimes called “Face blindness”
- Rely on non-facial cues (voice, gait, clothing)
- Recognize that they are looking at a face.
- Can discriminate some qualities (gender, race, ~age, emotional expression!)

Prosopagnosia

- Superior temporal sulcus has strong connections with the amygdala. Injury can cause a severe deficit in comprehension of facial expressions and gaze directions. Adolphs et al. 1994, Young et al., 1995
- Can be secondary to PCA infarct encroaching on medial cortex of occipital and temporal lobes and the inferior longitudinal fasciculus (i.e., occipital to temporal projections).
- Can be associated with Alzheimer's (and much else).
- Most strongly associated with the fusiform gyrus: in particular the *fusiform face area*.


Central sulcus
PARACENTRAL LOBULE
SUP FRONTAL GYRUS
PRAECUNEUS
Cingulate sulcus
CINGULATE GYRUS
Parieto-occip. fissure
CUNEUS
Corpus callosum
Fornix
Calcarine fissure
ISTHMUS
LINGUAL GYRUS
UNCUS
Parolfactory area
HIPPOCAMPAL GYRUS
Infr temp.
FUSIFORM GYRUS
INF. TEMP. GYRUS
sulcus

Central Achromatopsia

- Can perceive borders defined by color differences that classic photoreceptor color-blind people cannot (Ishihara test).
- Pure form is very rare (requires specific bilateral lesions in ventro-medial occipital lobe)
- A defect in color perception caused by an acquired cerebral lesion.
  - Reduced hue discrimination
  - Deficient color constancy
  - Cannot order colors in a test
- Colors are either all gray, or “washed out”, “dirty”, or “faded”
- Problems in imagining colors are reported (Oliver Sachs, colorblind painter)
- Also known as *color agnosia*

Balint Syndrome

- Simultanagnosia (independent of object size or visual periphery.)
- Spatial disorientation: inability to orient, localize, and reach for objects = “optic ataxia”
- Oculomotor disturbances: fixation, saccade initiation and accuracy, smooth-pursuit. “optic apraxia” = inability to voluntarily move eyes to a new position
- Damage to posterior superior watershed areas (Brodmann 7 + 19). Multiple strokes, Alzheimer’s, tumors, other injury.

Misidentification Syndromes

- Patient incorrectly identifies and reduplicates persons, places, objects, or events.
- Capgras Σ: multiple persons (incl. Family) replaced by impostors.
- Fregoli Σ: a familiar person is impersonating a stranger.
- Intermetamorphosis: Persons known to the patient have exchanged identities.
- Delusion of subjective doubles: the patient themselves have been replaced.

Capgras Syndrome

- Also called Capgras Delusion.
- Named after French psychiatrist Joseph Capgras, wrote a paper about a woman who was convinced that “doubles” had replaced people she knew.
- It has been proposed that face recognition has both conscious and unconscious parts. Idea that people can recognize the faces, but they associate them with strangers (don’t show appropriate emotional reactions to faces they recognize). Visual processing defect has to be in addition to other cognitive problems (schizophrenia). Not just cognitive: cognitive + visual processing.
- (some prosopagnosics show opposite: no recognition but appropriate affect, i.e. emotional response).

Fregoli Syndrome

- Also Fregoli Delusion
- Named after Italian actor and quick-change artist Leopoldo Fregoli
- Belief that different people are a single person in disguise.
- Like Capgras, thought to be a problem with normal face recognition.

Reduplicative paramnesia

- Typically misidentification of places rather than person. A delusional belief that a place or location exists in two places at once.
- Originally thought to be ‘psychological’, currently the thinking is that it involves disorders of visual spatial processing and memory (temporal cortex and hippocampus?).

The following excerpt illustrates some of the core features of the delusion. The patient had suffered a head injury after a fall in his home. The impact had caused a fractured skull and frontal lobe damage to both sides (although more pronounced on the right) owing to the formation of intracerebral hematomas:

“A few days after admission to the Neurobehavioural Center, orientation for time was intact, he could give details of the accident (as related to him by others), could remember his doctors' names and could learn new information and retain it indefinitely. He exhibited, however, a distinct abnormality of orientation for place. While he quickly learned and remembered that he was at the Jamaica Plain Veterans Hospital (also known as the Boston Veterans Administration Hospital), he insisted that the hospital was located in Taunton, Massachusetts, his home town. Under close questioning, he acknowledged that Jamaica Plain was part of Boston and admitted it would be strange for there to be two Jamaica Plain Veterans Hospitals. Nonetheless, he insisted that he was presently hospitalized in a branch of the Jamaica Plain Veterans Hospital located in Taunton. At one time he stated that the hospital was located in the spare bedroom of his house.” Benson et al 1976

Reduplicative paramnesia (RP), a syndrome named by Pick in 1903 to describe a specific and limited disturbance of memory, is characterized by a subjective certainty that a familiar place or person has been duplicated.1 Most often seen in posttraumatic brain injuries, this syndrome has been described in a variety of neurologic conditions, including strokes, intracerebral hemorrhages, tumors, dementias, encephalopathies, and various psychiatric conditions.2 The classic and most commonly described neuroanatomical localization is the combination of bilateral frontal lobe and right hemisphere lesions.3,4 Although many theories have been advanced to explain how these lesions could produce this syndrome, the specific pathophysiology is unknown. We report a patient who provides anatomical and neuropsychological support for the theory that RP is a syndrome that may develop in a vulnerable brain by a lesion in the ventral visual stream, disrupting communication between the visual cortex and both visual processing areas in the inferior temporal lobe and visual memory in the nondominant parahippocampal region.

From Budson et al., 2000.

Hemi-Neglect

- Cannot notice or pay attention to visual stimuli on one hemifield. Can't even imagine that half the world is even there! Not even black…
- Usually it's the left visual hemifield that is neglected ("Eyes Right"), due to a right parietal lesion. (The opposite almost never happens: the visual system no longer symmetrical at this level?).
- Relatively common

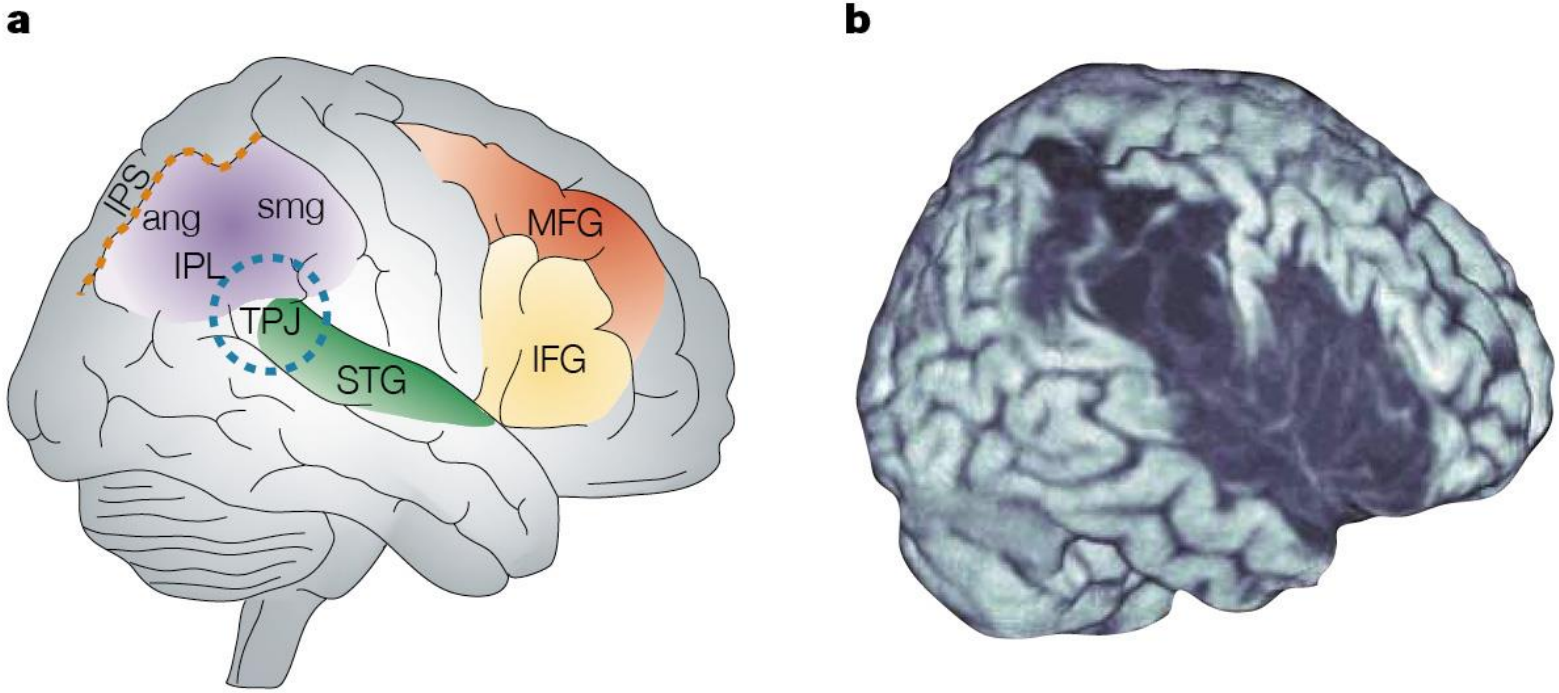


Figure 1 | **Neuroanatomy of neglect. a** | Cortical regions that are damaged in patients with neglect. Posterior areas include the junction of the temporal and parietal lobes (TPJ), the inferior parietal lobe (IPL) which includes the angular (ang) and supramarginal gyrus (smg), the intraparietal sulcus (IPS), and the more recently implicated superior temporal gyrus (STG). Frontal areas include the inferior frontal gyrus (IFG) and the middle frontal gyrus (MFG). **b** | Reconstruction of the cortical areas destroyed by stroke in a patient with neglect. In this patient, all the areas implicated in **a** are affected, but in other patients more focal damage, restricted to one or two of the areas shown in **a**, can occur.

From Husain and Rorden 2003

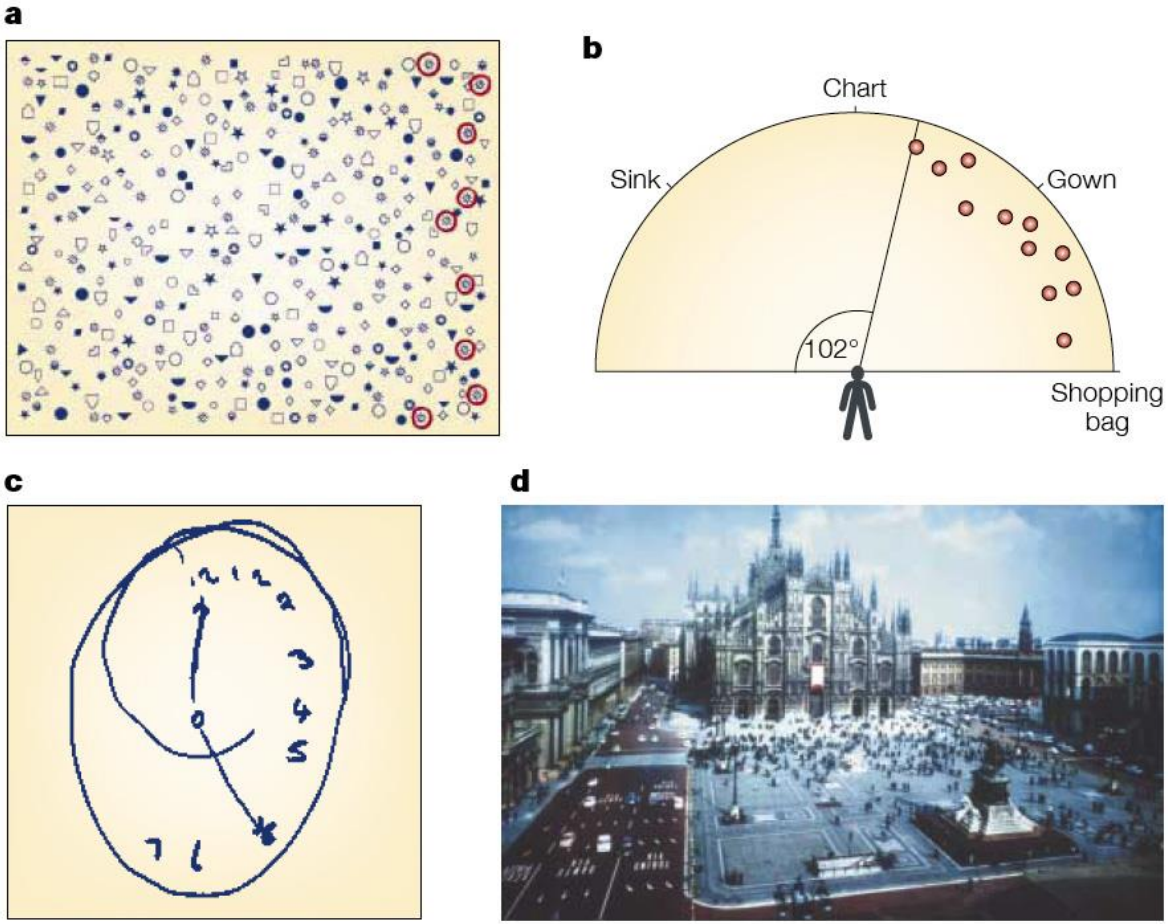


Figure 2 | **Clinical tests for neglect. a** | Cancellation test. The visual search of a typical left neglect patient is restricted to finding and marking targets on the ipsilesional (right) side, and missing those towards the contralesional (left) side. **b** | Naming objects around a room. This simple task reveals that only items to the right were reported. Modified, with permission, from REF. 3 © 1991 BMJ Publishing Group **c** | Clock drawing. Some patients with neglect omit details on the contralesional side of a drawing from memory, such as a clock. **d** | Milan Square. In a famous study, Milanese left neglect patients failed to recollect places on the left of the city's Piazza del Duomo when asked to imagine the square from one vantage point, but were able to report these locations when asked to imagine looking at the square from the opposite side.

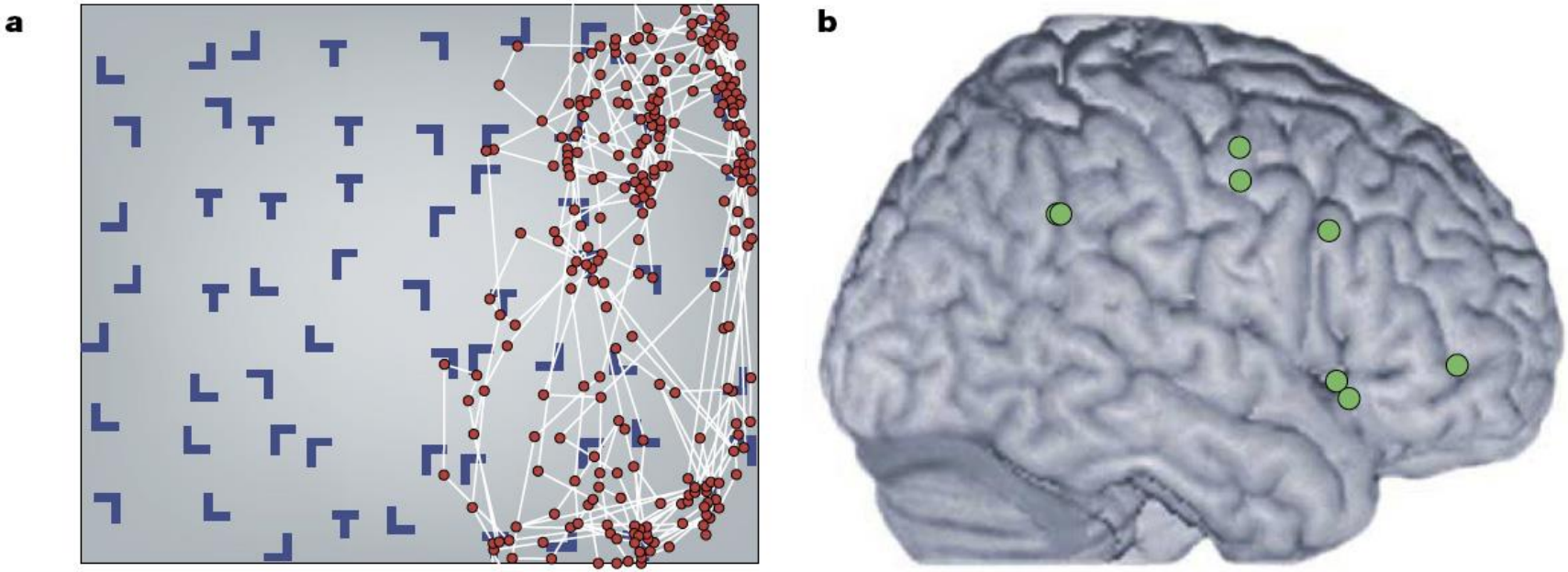
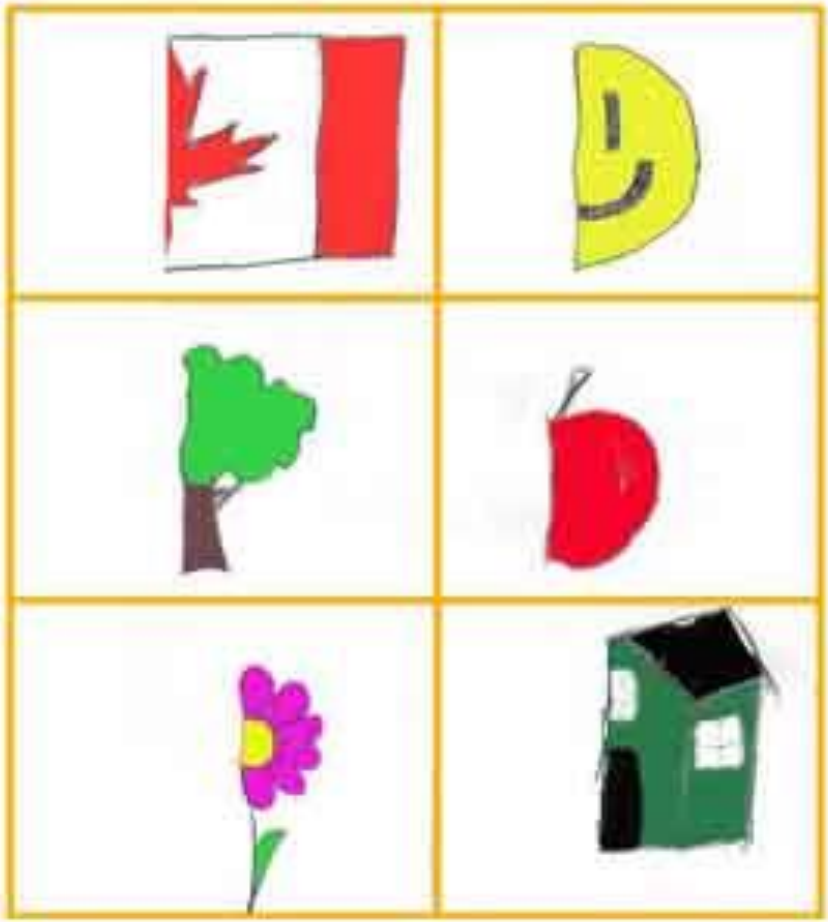


Figure 6 | **Trans-saccadic spatial working memory. a** | Eye movements of a right-hemisphere patient with neglect. The red dots show the fixation points (where the patient looks) while he searches for targets (Ts) among distractors (Ls). The white lines show his scan path. Note how he recursively revisits locations he has searched before. Modified, with permission, from REF. 43 © 2001 Oxford University Press. **b** | Green dots represent lateral parietal and frontal areas activated in double or triple memory-guided saccadic paradigms. These tasks are considered to require updating of spatial representations across saccades.

Illustrations of the type of errors in drawing objects (whether from memory or not) made by patients with visual-neglect (hemi-inattention, hemi-neglect) where the left half of space is not drawn; typically the damage is in the right hemisphere.



Riddoch Phenomenon

After George Riddoch, a British neurologist who first described the syndrome in 1917. Also called "statokinetic dissociation". This is the preservation of the perception of motion in an otherwise blind hemifield after a visual cortical (occipital) lesion.

The objects have no form or color, only motion can be perceived. Subjects describe the motion as being like "a black shadow moving on a black background" (so not really blindsight, where there is no conscious perception at all). Possibly caused by direct activation of MT via alternate pathways (maybe LGN to MT direct, or superior colliculus to MT), or by small 'islands' of remaining functionality in visual cortex.

Visual Ataxia

Patients with homonymous hemianopia due to occipital lobe lesions may experience loss of balance due to altered sensory input I.e. not a problem with cerebellum vestibular system etc. Possible sensation of falling towards the blind hemifield.

Akinetopsia

Loss of perception of motion with preservation of the perception of other modalities of vision (color, shape, etc). Damage to the human *homolog* of MT/V5 (“human MT” sometimes called MT+ or just MT).

Akinetopsia was mentioned in the 2006 episode of House titled "Son of Coma Guy." Dr. House diagnosed the patient by flashing the lights of the hospital room which caused the patient to have a seizure. The patient also could not catch a bag of chips and was startled when Dr. House moved closer to him as the motion was undetected until the motion ceased (wikipedia).

Visual Allesthesia

A disorder of visual perception where the visual fields appear to be tilted, rotated or flipped.

Anosognosia is a condition in which a person who suffers disability due to brain injury seems unaware of or denies the existence of his or her handicap. This may include unawareness of quite dramatic impairments, such as blindness or paralysis. It was first named by neurologist Joseph Babinski in 1914, although relatively little has been discovered about the cause of the condition since its initial identification. The word comes from the Greek words *"nosos"* disease and *"gnosis"* knowledge. (wikipedia)

Anton-Babinski syndrome, more frequently known as **Anton's blindness**, or **Anton's Syndrome**, is a symptom of brain damage occurring in the occipital lobe. People who suffer from it are "cortically blind," but affirm, often quite adamantly and in the face of clear evidence of their blindness, that they are capable of seeing. Failure to see is dismissed by the sufferer through confabulation. It is mostly seen following a stroke, but may also be seen after head injury. It's believed that the cause is damage to two areas; the portion of the brain responsible for eyesight, and the portion responsible for detecting the presence of vision. (wikipedia)

Hence, Anton's Syndrome includes anosognosia. Has also been reported to occur with optic nerve problems if frontal lobe damage also present.

Somatosensory Agnosias 體感覺認識不能

Astereognosia觸覺失認症 – inability to recognize objects by touch無法經由觸覺辨認物體

- pure cases are rare – other sensory deficits are usually present 單純的個案是稀少的- 其它感覺缺陷較常見

Asomatognosia 自體感覺失認症– the failure to recognize parts of one's own body 無法辨認部分自我個體的部分(the case of the man who fell out吵架 of bed)

Perception of Pain

- Despite its unpleasantness, pain is adaptive and needed 痛覺是適應性的、必需的
- No obvious cortical representation of pain 痛覺沒有顯著的皮質表現(although the anterior cingulate gyrus appears involved in the emotional component of pain雖然前扣帶回顯示了包含痛覺的情緒元素，但在不同研究、不同個案身上則有不同的位置表現)
- Descending pain control 痛覺下降控制–pain can be suppressed by cognitive and emotional factors 痛覺可以被抑制，經由認知及情緒因素

Descending Pain Control 減輕疼痛控制

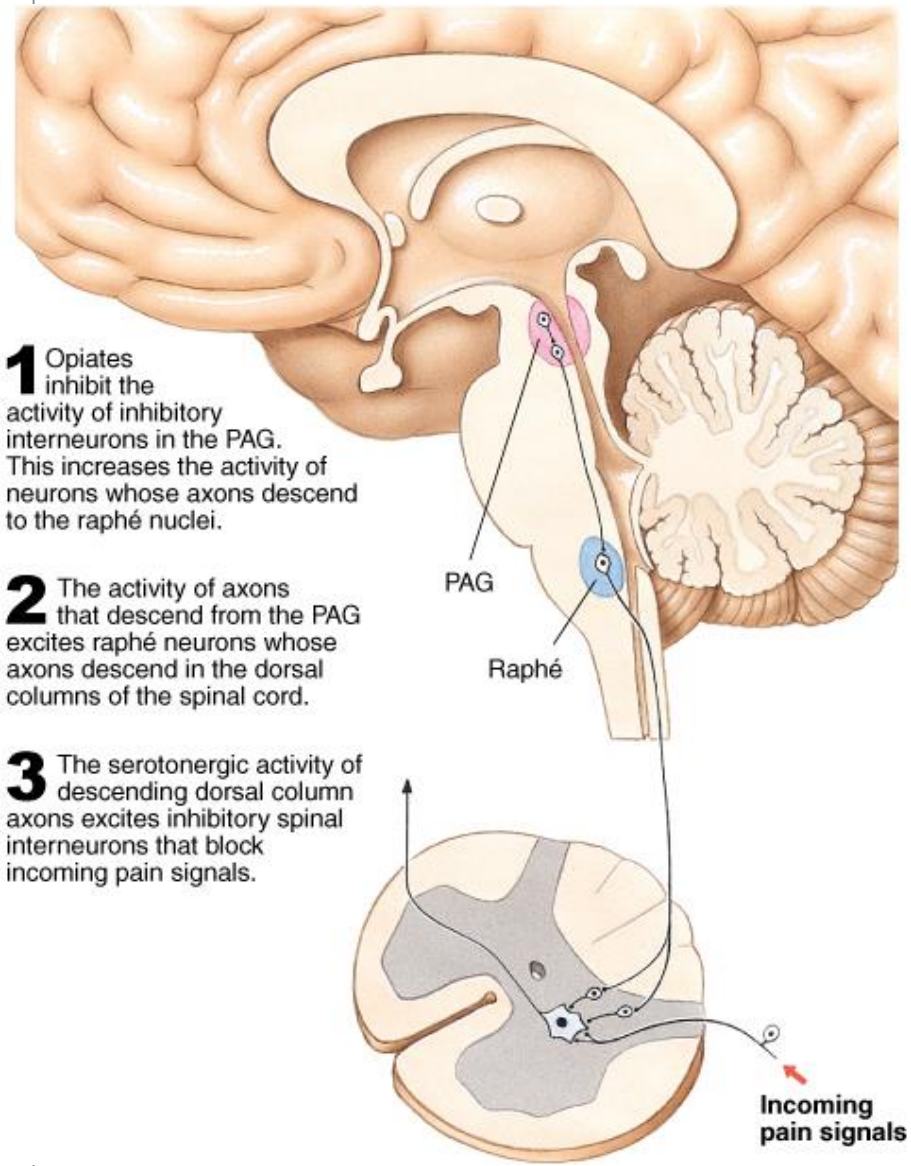


Figure 7.18: Basbaum and Fields's (1978) model of the descending analgesia circuit

- Circuitry迴路 identified by the following studies:
  - Electrical stimulation of the periaqueductal gray (PAG) has analgesic止痛 effects
  - PAG and other brain areas have opiate receptors鴉片受器
  - Existence of endogenous opiates (natural analgesics天然的止痛劑) – endorphins腦內啡

Neuropathic Pain 神經徑路疼痛

- Severe強烈的 chronic pain in the absence of a recognizable pain stimulus 缺乏可辨認的疼痛刺激源於強烈的慢性疼痛中
- Likely from pathology of nervous system linked to an injury 在治癒後依然會感到疼痛，如同神經系統聯結到一個傷害的病理機制
- Some evidence for aberrant glial cell signals triggering neural pain pathways 存在一些證據，指出神經膠細胞訊號會引起神經疼痛路徑的反應

Chemical Senses: Smell and Taste 化學感創: 嗅覺、味覺

Olfaction (smell) 嗅球

- Detects airborne chemicals
  偵測空氣中的化學元素

Gustation (taste) 味蕾

- Responds to chemicals in the mouth

  對舌頭上的化學反應

Food acts on both systems to produce flavor食物在兩個系統之間會產生“滋味“

Chemical Senses: Smell and Taste

- Pheromones are chemicals that influence that behavior of conspecifics (members of the same species) 費洛蒙（化學物質）會影響同一物種的行為
- Evidence of human pheromones人類費洛蒙的證據
- Changes in olfactory sensitivity across and menstrual cycle 嗅覺敏銳度的改變及月經週期
- Synchronization of menstrual cycles月經週期同步
- Sex identification by smell (especially by women) 由嗅覺辨認性別（特別是女人）
- Men can identify menstrual stage by smell 男性可經由嗅覺辨認月經的階段

Olfactory System

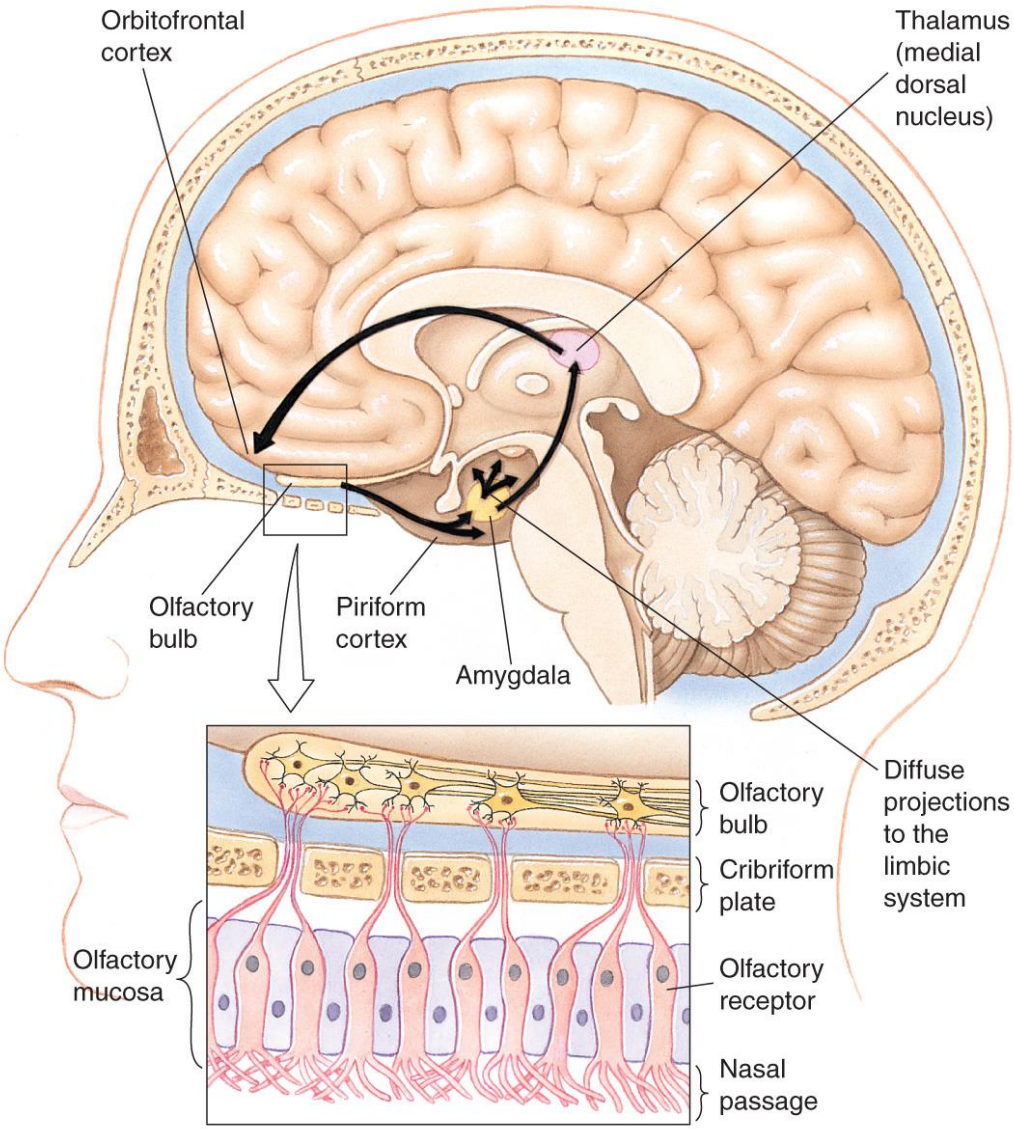
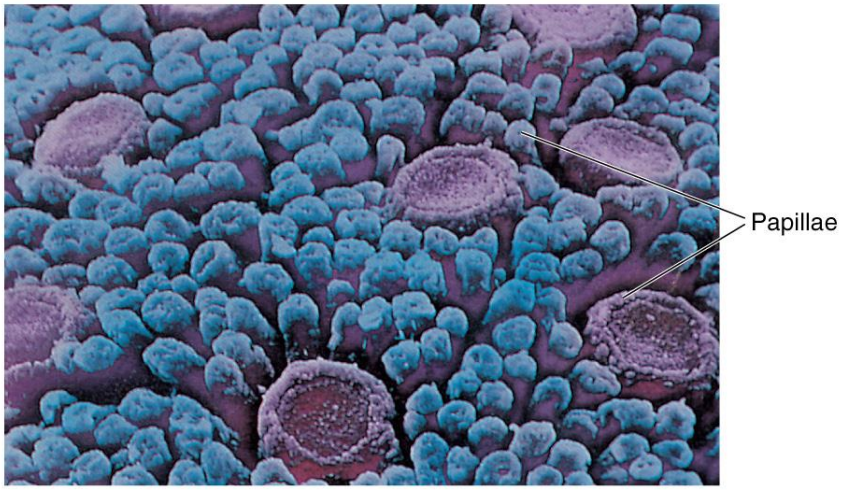


Figure 7.19: The human olfactory system

Gustatory System

Surface of Tongue



Cross Section of a Papilla

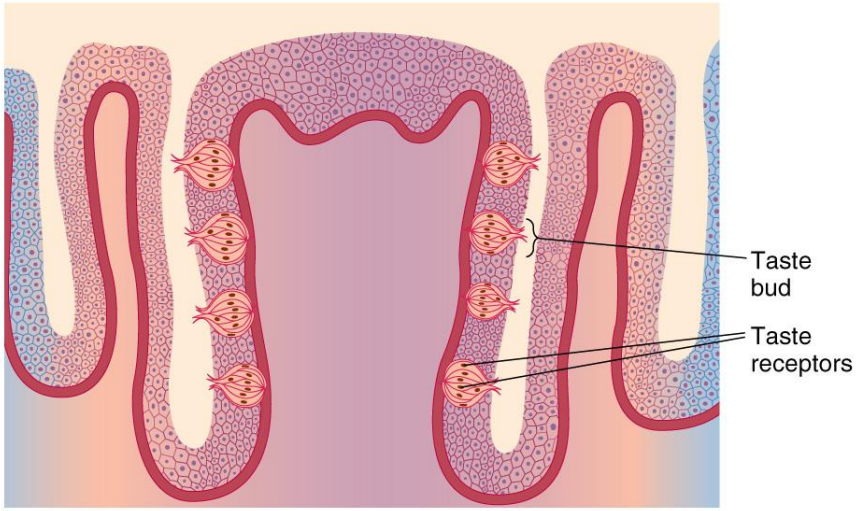


Figure 7.20: Taste receptors, taste buds, and papillae on the surface of the tongue

- **Receptors in tongue and oral cavity in clusters of about 50 called taste buds 舌頭上的受器及口腔裡近五十個叢集稱為味蕾(located around small protuberances called papillae位於舌頭週圍小小凸起稱為乳凸的地方)**
  - > 4 (sweet甜, sour酸, salty鹹, bitter苦) primary tastes – 5th is *umami*, meat or savory 鮮味(獨特的、有肉汁味道的、或是美好滋味的味覺)
  - Many tastes not created by combining primaries許多味覺的產生並非由上述幾種味覺混合產生
  - Salty and sour don't have receptors; they merely act on ion channels鹹跟酸並沒有受器;他們僅僅只是離子通道上的作用

Senses

Anosmia嗅覺喪失 – inability to smell

- Most common cause is a blow to the head that damages olfactory nerves 常見的導因是由於腦部撞擊而傷害到嗅覺神經
- Incomplete deficits seen with a variety of disorders 其它疾病也可能導致這個疾病

Ageusia味覺喪失– inability to taste

- Rare due to multiple pathways carrying taste information 少見，主因為由多種路徑承載味覺訊息

Selective Attention

Has 2 characteristics:

- **Improves perception of what is attended to with that which is not只注意到某些特定訊息**
- **Internal cognitive processes (endogenous attention) and external events (exogenous attention) focus attention. 內在的認知歷程（內在的注意力）及外在事件（外在的注意力）**

Cocktail-party phenomenon 雞尾酒效應:

Indicates that there is processing of information not attended to 存在一個訊息處理歷程是我們所沒有注意到的

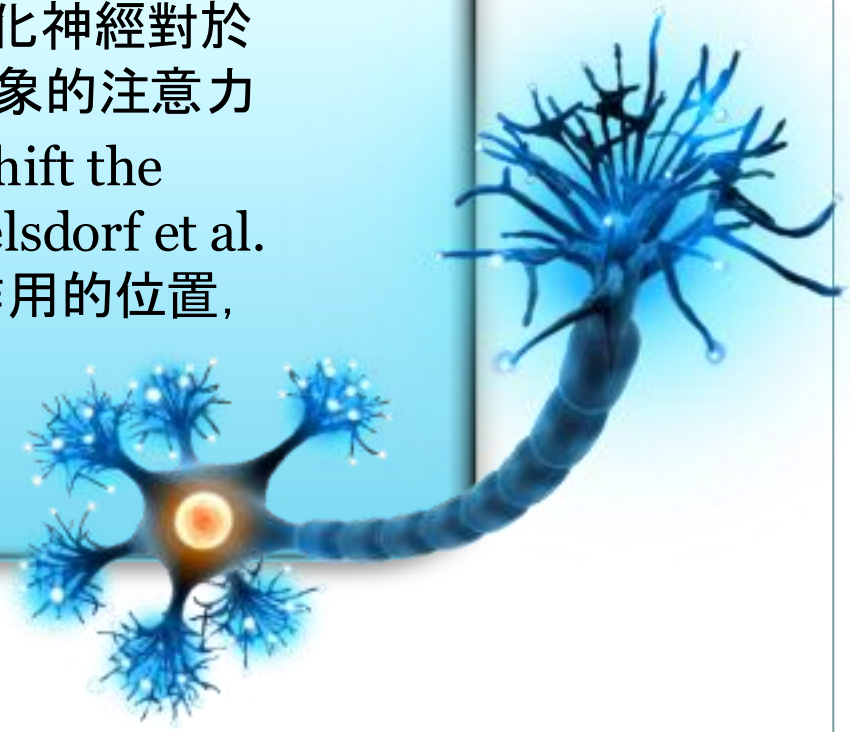
Change Blindness

Change blindness改變盲目 – no memory of that which is not attended to 對眼前發生的改變沒有注意到，沒注意到的訊息不會產生記憶

• We do not appear to remember parts of a scene that are not the focus of our attention 我們並未記得我們未注意到的場景部分

Neural Mechanisms of Attention

- Selective attention is thought to work by strengthening the neural responses to attended-to aspects and by weakening the responses to other 選擇性注意力被認為是經由強化神經對於注意對象的反應，經由弱化對其它對象的注意力
- For example, spatial attention can shift the location of receptive fields (Wommelsdorf et al. 2006) 空間性注意力可以轉移受器作用的位置，神經並非是靜止不動的



Simultanagnosia

Simultanagnosia 注意力障礙– a difficulty attending to more than one visual object at a time 無法在一個時間注意超過一個視覺物件

- Typical cause: bilateral damage to the dorsal stream背流（從V1傳到V2，再到背內區、V5中顳葉、及後頂葉皮質）兩側傷害 (involved with localizing objects in space包含了定位物件空間)